

Employee Attrition Prediction – Executive Summary

Project Summary

The objective of this project was to identify employees who are more likely to leave the company by analyzing employee information such as job role, department, monthly income, work-life balance, business travel, overtime, and years of experience. The goal is to help the HR department implement preventive measures to improve employee retention and reduce the costs associated with employee turnover.

The analysis showed that the **Sales department** has the highest employee attrition rate, with **Sales Representatives** experiencing the greatest turnover. Employees who frequently work **overtime** and those who **travel frequently for business** were also found to have a higher likelihood of leaving the company. In addition, most employee exits occur during the **first five years** of employment, indicating that new employees require greater support and engagement.

Although employees with lower monthly incomes were more likely to leave, the analysis showed that **salary is not the only factor** influencing attrition. Work-life balance, overtime, business travel, and job role all have a significant impact on employee retention.

Based on these findings, it is recommended that the company reduce excessive overtime, improve work-life balance initiatives, and strengthen employee engagement programs, particularly for employees in the Sales department and those who travel frequently. Regular career development discussions, mentoring programs, and early retention initiatives during the first few years of employment may further reduce attrition.

This predictive model can support HR teams in identifying employees who may be at risk of leaving. However, it should be used as a decision-support tool alongside human judgment, as personal circumstances and external factors that influence employee decisions are not captured in the available data.